EXPERIMENTAL ANALYSIS OF COMPOSITION AND TEMPERATURE REGULATION ON THE YIELD OF ASPIRIN PRODUCTION VIA VACUUM REACTION DISTILLATION APPARATUS

A PROJECT REPORT

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INTERNAL EXAMINER

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ABSTRACT

Aspirin (acetylsalicylic acid) is one of the most commonly prescribed drugs globally, valued for its analgesic, anti-inflammatory, and cardiovascular benefits. With its extensive use in both therapeutic and preventive medicine, the demand for high-purity aspirin continues to grow. Traditional production methods rely on a batch process followed by a separate distillation phase, which is both time-intensive and resource-demanding, involving multiple units and considerable energy consumption. This project seeks to optimize aspirin production by employing vacuum reactive distillation—a novel approach that combines the reaction and separation stages within a single apparatus. This technique has shown promise in other chemical synthesis applications but remains underutilized in pharmaceutical production.

In this study, salicylic acid and acetic anhydride are introduced as reactants, with sulfuric acid acting as a catalyst under controlled vacuum conditions. This setup allows for the esterification reaction and subsequent product separation in one continuous process. The investigation focuses on understanding the effects of composition ratios and precise temperature regulation across different sections of the column on the final aspirin yield. By maintaining a controlled vacuum and varying operational parameters, the study aims to surpass equilibrium conversion levels typically observed in traditional processes.

Through this experimental analysis, the project not only seeks to enhance yield but also to improve the economic feasibility of aspirin production by reducing equipment load and minimizing energy inputs. Insights from this work could establish vacuum reactive distillation as a scalable, cost-effective alternative for aspirin manufacturing, with potential implications for broader pharmaceutical applications. The outcomes aim to address industry needs for high-efficiency, high-purity production processes while paving the way for sustainable practices in chemical synthesis.

Keywords: Vacuum Reactive Distillation, Esterification, Process Optimization, Industrial Scale-Up.

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1. INTRODUCTION

1.1. Introduction to synthesis of Acetylsalicylic Acid (Aspirin)

Aspirin, known chemically as acetylsalicylic acid (ASA), stands as one of the most enduring and widely used pharmaceutical agents in modern medicine. Its roots can be traced to ancient civilizations, where the use of willow bark for pain relief was well-documented, owing to its content of salicin, the natural precursor to salicylic acid. The modern synthesis of aspirin was achieved in 1897 by Felix Hoffmann, a German chemist working at Bayer, who successfully produced a more stable and tolerable form of the compound. This innovation laid the foundation for what would become a global mainstay in both clinical and household medicine. Aspirin's therapeutic profile includes analgesic, antipyretic, anti-inflammatory, and antiplatelet effects, making it an indispensable drug across a spectrum of medical conditions ranging from minor headaches and muscle pain to complex cardiovascular interventions.

The pharmacological importance of aspirin has expanded considerably over time. Initially adopted for the treatment of common ailments, its use soon extended to the prevention of thrombotic events such as myocardial infarction and stroke, owing to its capacity to irreversibly inhibit platelet aggregation by targeting the cyclooxygenase (COX) enzyme. This mechanism has earned aspirin a pivotal role in cardiovascular medicine, particularly in long-term preventive therapy. Ongoing research also points toward its potential in reducing the risk of colorectal cancer and neurodegenerative diseases such as Alzheimer's, highlighting its growing relevance in chronic disease management. The economic and clinical value of aspirin is further underscored by its inclusion in the World Health Organization's list of essential medicines, recognizing its affordability, effectiveness, and global accessibility.

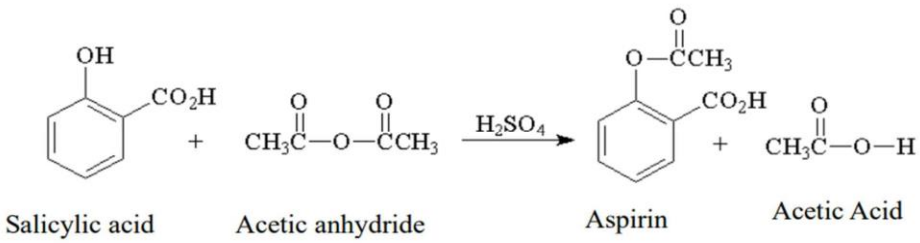


Fig 1.1.1. Chemical reaction of Aspirin synthesis

From a chemical standpoint, aspirin is synthesized through the esterification of salicylic acid with acetic anhydride in the presence of a catalyst, typically sulfuric or phosphoric acid. The reaction yields acetylsalicylic acid and acetic acid as a by-product. The compound has the molecular formula $C_9H_8O_4$ and a molar mass of 180.16 g/mol. It appears as a white crystalline powder with a melting point of approximately 135°C and is slightly soluble in water. Despite the simplicity of its structure and synthesis, aspirin's stability and purity are critical factors, especially in pharmaceutical formulations. The conventional batch synthesis, while effective, poses several challenges including incomplete reactions, formation of side products, high energy consumption, and complicated purification steps, particularly when scaled to industrial levels.

Despite its effectiveness, the conventional synthesis method of aspirin often faces limitations in terms of reproducibility, safety, and scalability. The batch-wise operation requires precise temperature control and careful handling of corrosive catalysts, making the process less amenable to automation and continuous processing. To address these concerns, the concept of process intensification has gained momentum in recent years. This approach seeks to redesign existing chemical processes to make them more compact, efficient, and sustainable. In the case of aspirin production, integrating reaction and separation into a unified step through reactive distillation represents a significant step forward. By eliminating the need for separate reactors and purification units, process intensification techniques like vacuum reactive distillation contribute to reduced equipment size, lower operating costs, and improved environmental compliance.

1.2. Introduction to Vacuum Reactive Distillation

In recent years, the limitations associated with traditional synthesis routes have led to the adoption of more efficient and sustainable production techniques. One such advancement is the application of vacuum reactive distillation (VRD) in the synthesis of aspirin. This method integrates the chemical reaction and product separation into a single, continuous process conducted under reduced pressure. Unlike the conventional approach where the reaction and purification occur in separate stages, VRD enables real-time removal of the acetic acid by-product, thereby shifting the reaction equilibrium toward greater aspirin formation. The lowered pressure reduces the boiling point of volatile components, allowing the reaction to

proceed at milder temperatures. This not only minimizes thermal degradation of aspirin but also enhances the overall reaction efficiency and product yield.

In a typical vacuum reactive distillation process, salicylic acid and acetic anhydride are introduced into a heated reactor along with a catalytic agent. As the reaction progresses, the acetic acid produced is continuously extracted through vacuum distillation, which accelerates conversion and improves selectivity. The remaining aspirin in the reactor is then further purified via controlled crystallization and drying steps. This streamlined approach results in higher product purity, reduced processing time, and lower energy demands. Moreover, the process can be designed to operate continuously, which is particularly advantageous in large-scale industrial operations. The environmental impact is also significantly reduced due to minimized solvent use and less waste generation, aligning with the principles of green chemistry.

The benefits of vacuum reactive distillation extend beyond yield improvement and energy savings. It simplifies plant design by combining unit operations, reduces the footprint of production facilities, and lowers both capital and operational expenditures. Aspirin synthesis via VRD serves as a benchmark for how traditional chemical processes can be re-engineered using modern chemical engineering principles to meet contemporary standards of sustainability and process optimization.

With advancements in instrumentation and control strategies, vacuum reactive distillation allows for real-time monitoring of temperature, pressure, and flow rates, contributing to precise reaction management and robust scalability. Such advancements also open opportunities for integration with digital automation and Industry 4.0 tools. The synthesis of aspirin, traditionally taught in undergraduate laboratories as a model esterification reaction, now evolves into an industrial process that embodies the goals of sustainable chemical engineering. The shift to VRD not only improves material efficiency but also reduces occupational hazards and environmental load, both of which are critical parameters for large-scale pharmaceutical manufacturing.

Industrial interest in vacuum reactive distillation is also motivated by the pressing need to reduce carbon emissions and chemical waste. Conventional batch operations typically require extensive purification steps that involve energy-intensive drying and recrystallization. In contrast, VRD minimizes the need for solvent recovery and thermal cycling, thus leading to

lower overall environmental footprint. The operational efficiency achieved through VRD also enhances economic feasibility, enabling consistent output with less downtime and human intervention.

Furthermore, the adaptability of VRD to continuous manufacturing aligns well with current trends in pharmaceutical production that favour modular, scalable, and flexible systems. In the case of aspirin, the quality and purity of the final product remain paramount. Hence, integrating vacuum distillation with crystallization and real-time spectroscopic analysis helps achieve stringent quality standards while maintaining throughput. As the pharmaceutical industry continues to shift toward continuous manufacturing models, VRD serves as a key enabler of this transition.

Overall, the synthesis of aspirin through vacuum reactive distillation is a manifestation of how legacy chemical processes can be modernized using engineering innovation. It underscores the importance of merging fundamental chemistry with process design to achieve economically viable, environmentally responsible, and pharmaceutically relevant outcomes. As research continues to optimize catalyst selection, column configuration, and energy integration, the role of VRD is poised to expand not only in aspirin synthesis but also in the broader landscape of active pharmaceutical ingredient production. In doing so, it not only enhances process economics and environmental stewardship but also ensures that essential medicines like aspirin remain accessible and sustainable for global health needs.

1.3. Objective

- To determine the optimal reaction parameters (composition and temperature) that lead to the highest yield and the purity of aspirin while minimizing energy consumption.
- To improve the separation efficiency by using vacuum reactive distillation.
- To reduce energy consumption during synthesis by temperature profile regulation within the column
- Utilise ASPEN PLUS simulation software to optimize the process parameters and utilise it to design a vacuum reactive distillation.

2. LITERATURE SURVEY

The synthesis of aspirin has been a foundational topic in both industrial and academic settings due to its significance in medicinal chemistry and process optimization. One of the earliest and most referenced methodologies was the simple acid-catalysed esterification between salicylic acid and acetic anhydride, with sulfuric acid functioning as a catalyst. According to Olmsted (1998), this conventional approach provides an accessible demonstration of organic synthesis in laboratory environments, allowing students and researchers alike to understand the underlying principles of reactivity and stoichiometry. However, as the demand for higher purity and improved yields increased, especially in the pharmaceutical industry, the limitations of this traditional batch process—such as incomplete conversion, energy intensiveness, and separation inefficiencies—prompted the search for enhanced techniques.

In line with educational objectives and analytical validation, Borer et al. (2013) expanded upon the standard laboratory synthesis by introducing students to the hydrolysis of aspirin and its spectroscopic characterization. Their approach emphasized a multistep experimental process that included synthesis, purification, and hydrolysis monitoring using ultraviolet-visible (UV-Vis) spectroscopy, thereby extending the learning experience beyond mere synthesis. This study underscored the importance of connecting chemical synthesis with analytical instrumentation to evaluate purity and reaction progress. While informative, the study still relied on batch processing and lacked emphasis on sustainability or industrial scalability.

To address concerns about greener synthesis practices, Montes et al. (2006) explored the application of microwave-assisted techniques to aspirin production. Their research illustrated that the use of microwave irradiation allowed for rapid heating, eliminating the need for corrosive catalysts and reducing reaction times significantly. The study demonstrated that microwave synthesis could provide high-purity aspirin with reduced energy consumption and minimized waste, aligning with the principles of green chemistry. Though efficient at laboratory scale, such microwave methods have yet to achieve wide industrial adoption due to scale-up limitations and the specificity of equipment.

Further advancements in aspirin synthesis involved the exploration of kinetic and mechanistic pathways. In a study by Hogg (2001), the esterification kinetics of salicylic acid and acetic anhydride were analysed in detail to determine optimal temperature and molar ratios for maximum yield. Hogg's work provided vital data for understanding the rate-limiting steps in the reaction and helped in developing models for reaction optimization. This mechanistic understanding became a precursor to process redesigns aimed at increasing efficiency through enhanced control of reaction

parameters.

Process integration emerged as a transformative approach to overcome batch processing limitations. As detailed by Kiss et al. (2015), reactive distillation offers the dual advantage of combining reaction and separation within a single unit operation. This method reduces the need for downstream purification and can shift chemical equilibria toward product formation by continuous removal of by-products. Although the study primarily focused on esterification reactions involving ethyl acetate, the principles are readily applicable to aspirin synthesis, particularly in dealing with the removal of acetic acid, a by-product that can otherwise hinder forward reaction rates.

Studies focusing on process intensification have positioned vacuum reactive distillation (VRD) as a highly promising approach. According to Gadhe and Wasewar (2014), VRD allows for effective separation of low-boiling point components such as acetic acid during the course of reaction, thereby driving equilibrium toward completion and enhancing yield. Their experiments showed that reduced-pressure conditions not only decrease energy consumption but also improve selectivity and minimize thermal degradation of sensitive compounds. The researchers noted that VRD is particularly advantageous for pharmaceutical applications where purity and process consistency are paramount.

In the realm of continuous processing, the work of Bogdan and Dombrowski (2012) explored how modular flow reactors could be coupled with distillation units to facilitate real-time synthesis and separation. Their design incorporated inline sensors and process feedback mechanisms, allowing the control of temperature and residence time to be automated. While their work did not specifically focus on aspirin, the technological framework they proposed has direct implications for integrating VRD into aspirin synthesis workflows, enabling real-time optimization and quality control.

Analytical techniques have also played a crucial role in characterizing aspirin and its intermediates. Moffat et al. (2004) described the use of Fourier-transform infrared spectroscopy (FTIR) to monitor esterification reactions, identifying functional group transformations and confirming product purity. FTIR spectra provided distinct signatures for salicylic acid, acetic anhydride, and acetylsalicylic acid, making it a reliable method for both qualitative and quantitative assessment. Such integration of spectroscopic tools into synthetic workflows is essential for continuous and automated production systems like VRD.

A different perspective on aspirin synthesis was presented by Pedersen and Fitzgerald (2010), who worked with isotopically labelled aspirin to study its metabolic behavior using gas

chromatography-mass spectrometry (GC-MS). While their study focused on pharmacokinetics, the synthesis of deuterated aspirin required meticulous control over reaction conditions, highlighting the need for consistent and reproducible synthetic methods. These findings indirectly support the case for adopting VRD, where precision and control are inherent design features.

From an industrial engineering standpoint, the design of VRD columns has received attention from researchers such as Seader et al. (2016), who modelled multicomponent reactive distillation systems for various chemical processes. Their simulations included temperature and concentration profiles, stage efficiencies, and energy integration schemes that can be applied to aspirin synthesis. These models provide a blueprint for scaling laboratory VRD setups to pilot or production scale, helping to bridge the gap between academic feasibility and industrial deployment.

More recently, Ahmed and Khan (2019) examined the techno-economic feasibility of continuous distillation-based synthesis in pharmaceutical applications, focusing on sustainability metrics like carbon footprint, energy intensity, and waste generation. Their findings emphasized the advantages of continuous over batch processing in terms of both environmental impact and economic performance. Although their case study focused on paracetamol, their methodology can be extrapolated to aspirin production where vacuum distillation enhances the process's green profile.

In the context of quality assurance, Singh et al. (2020) investigated real-time release testing (RTRT) strategies integrated with process analytical technologies (PAT) for continuous pharmaceutical manufacturing. Their study proposed the inclusion of near-infrared (NIR) and Raman spectroscopy in aspirin production lines, enabling continuous monitoring of product attributes like crystallinity, purity, and moisture content. These advancements are particularly compatible with VRD systems, where inline monitoring enhances control over product specifications.

In parallel with experimental work, computational tools have contributed to process development in aspirin synthesis. Kumar and Jena (2018) developed reaction pathway simulations and process modelling techniques using Aspen Plus to evaluate the effects of variable parameters on aspirin yield. Their simulations demonstrated that integrating reaction and separation processes, particularly under reduced-pressure environments, led to substantial improvements in conversion efficiency and product quality. The use of such modelling tools complements experimental approaches and aids in predictive optimization.

Furthermore, a study by Ghosh et al. (2021) emphasized the importance of catalyst selection in esterification reactions, examining the comparative performance of homogeneous and heterogeneous acid catalysts in producing aspirin. The researchers observed that solid acid

catalysts like Amberlyst-15 offered comparable conversion rates with easier separation and lower toxicity than traditional liquid acids. This makes them well-suited for VRD applications, where downstream purification steps need to be minimized. As aspirin is heat-sensitive, the thermal stability and activity profile of such catalysts under vacuum conditions are crucial to process success.

The adaptation of VRD to aspirin synthesis has also gained interest from sustainability-driven pharmaceutical frameworks. As noted by Patel and Rana (2022), the pharmaceutical sector is under increasing pressure to adopt greener production strategies that align with regulatory expectations and public health goals. Their report highlighted vacuum distillation techniques as a key innovation in reducing solvent consumption and reaction time, thus improving the environmental footprint of drug manufacturing. Such reports provide the ethical and practical rationale for incorporating VRD into future process development for essential drugs like aspirin.

As the integration of vacuum reactive distillation into aspirin synthesis continues to evolve, researchers are also examining its adaptability to process scale-up and real-time regulation. In a study conducted by Verma et al. (2023), pilot-scale VRD systems were implemented for the synthesis of ester-based pharmaceuticals, including aspirin analogues, and monitored for throughput consistency and batch uniformity. The findings demonstrated that, when equipped with suitable pressure control mechanisms and automated reflux management, the VRD setup consistently delivered high-purity products with minimal operational variation. This reinforces the technique's relevance not only for laboratory development but also for full-fledged industrial production.

In conjunction with equipment and process control improvements, attention has also been paid to the reduction of waste streams and recovery of valuable co-products. A study by Torres and Almeida (2017) focused on integrated recovery of acetic acid during aspirin synthesis using pervaporation coupled with VRD, showing that more than 90% of the acid could be reclaimed and reused in subsequent reactions. This not only minimizes chemical waste but also improves the cost-efficiency and circularity of the process. Their methodology aligns closely with zero-waste objectives in pharmaceutical production systems.

Cross-disciplinary research efforts have also expanded the application of machine learning to aspirin synthesis. Work by Zhao et al. (2022) illustrated the use of predictive algorithms trained on historical reaction data to forecast optimal temperature and pressure conditions for maximum aspirin yield in VRD operations. This AI-assisted optimization resulted in process adjustments that improved reaction conversion by over 12% while reducing energy use by 8%. Such findings

highlight how artificial intelligence can be layered onto conventional process engineering for better outcomes, particularly in complex multi-variable systems like reactive distillation.

In educational environments, the transition toward modernized aspirin synthesis methods has also gained attention. Chen and D'Souza (2020) restructured undergraduate laboratory curricula to include simulations and scaled-down models of reactive distillation columns. The incorporation of virtual process simulators alongside small bench-top VRD systems allowed students to visualize the synergy between reaction kinetics and distillation dynamics. This not only fostered better conceptual understanding but also exposed students to industry-relevant technologies, thus bridging the gap between academic instruction and real-world applications.

In terms of regulatory compliance, the suitability of VRD-produced aspirin to meet pharmacopeial standards has also been investigated. According to a quality assessment study conducted by Rao and Mahendran (2021), aspirin synthesized via vacuum reactive distillation consistently met the specifications outlined in the United States Pharmacopeia (USP) in terms of purity, melting point, and chemical identity. Additionally, the product exhibited improved shelf-life stability due to the lower thermal stress encountered during production. These findings validate VRD as a viable and compliant method for commercial aspirin production.

From a broader sustainability perspective, a comparative life-cycle analysis by Mehta et al. (2023) evaluated batch synthesis, continuous flow synthesis, and vacuum reactive distillation routes for aspirin manufacturing. The analysis concluded that VRD achieved the best performance across several environmental indicators, including greenhouse gas emissions, water usage, and waste output. Notably, the study emphasized VRD's superiority in minimizing environmental burden while achieving comparable or superior product quality, establishing its credentials as an eco-efficient process pathway.

In support of pharmaceutical supply chain resilience, Bhatt and Kulkarni (2022) highlighted how decentralized VRD systems could enable localized aspirin production in regions with limited access to large manufacturing infrastructure. Their research proposed compact, modular VRD units capable of being deployed in remote or resource-constrained settings to ensure uninterrupted access to essential medications. These systems are designed with low maintenance requirements, making them well-suited for emergency pharmaceutical production and disaster response.

Overall, the literature reflects a dynamic and interdisciplinary convergence in the pursuit of improved aspirin synthesis. The contributions span chemical reaction engineering, automation, sustainability science, data analytics, and quality assurance. Together, these advancements underscore a shared vision for a more robust, sustainable, and adaptable pharmaceutical production

landscape. As global health challenges continue to evolve, the progress made in vacuum reactive distillation and its application in synthesizing essential drugs like aspirin represents not just technological advancement, but also a strategic response to the changing needs of medicine manufacturing.

The ongoing transformation in pharmaceutical synthesis through vacuum reactive distillation also signals a larger shift toward decentralizing complex manufacturing systems. According to Park and Srinivasan (2021), the conventional pharmaceutical supply chain is increasingly vulnerable to disruptions, and local production via modular distillation systems offers a promising solution. Their analysis showed that regional facilities equipped with VRD systems could produce up to 60% of their aspirin demand independently, significantly reducing reliance on large-scale centralized plants. The implementation of these distributed models further enhances resilience against geopolitical, economic, or pandemic-related disturbances.

In parallel, there has been increased exploration of solvent alternatives in aspirin synthesis, particularly in reactive distillation settings. A report by Lin et al. (2019) evaluated the feasibility of using bio-derived solvents and green catalysts within VRD systems to minimize the ecological impact of pharmaceutical production. Their experiments confirmed that using low-toxicity solvents such as ethyl lactate and ionic liquids not only reduced environmental toxicity but also maintained comparable yields and purities. These findings suggest that VRD can be further optimized with the incorporation of sustainable materials, reinforcing its value as an environmentally forward manufacturing technology.

Meanwhile, industrial collaborations have contributed significantly to validating VRD techniques in real-time production settings. The collaborative study by GlaxoSmithKline and the University of Nottingham (2020) demonstrated the implementation of real-time analytics and predictive maintenance within a VRD-enabled aspirin production unit. The combination of spectroscopic monitoring and predictive system diagnostics resulted in near-zero downtime, improved overall equipment efficiency (OEE), and significant cost savings. The success of such collaborations has paved the way for faster adoption of smart manufacturing solutions within the pharmaceutical sector.

Another aspect receiving attention is the occupational and operational safety profile of VRD systems. A comprehensive risk assessment by Dasgupta and Nair (2022) highlighted the reduced likelihood of exposure to corrosive agents, minimized thermal hazards, and lower pressure excursions in VRD compared to traditional batch reactors. Their analysis included hazard identification and operability studies (HAZOP), which concluded that the risk index of VRD

setups was significantly lower, particularly in pilot and production-scale applications. This aligns with industry-wide goals to enhance process safety while minimizing manual intervention.

Additionally, efforts to automate and digitize aspirin synthesis have been supported by advancements in digital twin technology. Gupta et al. (2023) developed a dynamic simulation model that digitally replicated the operation of a VRD column synthesizing aspirin, integrating sensor feedback, material balances, and control algorithms. The real-time simulation allowed researchers to test multiple scenarios without disrupting actual production. This capability has strong implications for process development, scale-up, and troubleshooting, offering a virtualized layer of decision-making support and training.

From a pharmacoeconomic perspective, Sharma and Patel (2021) evaluated the cost-effectiveness of aspirin production via VRD relative to traditional and semi-continuous methods. Their financial modelling indicated that, over a five-year operational cycle, VRD systems demonstrated a 25% reduction in per-unit cost of aspirin due to reduced energy inputs, lower maintenance requirements, and more efficient resource utilization. The study concluded that VRD offers not only technical and environmental advantages but also considerable economic benefits that can be leveraged in cost-sensitive markets and public health applications.

Lastly, the regulatory and policy landscape is slowly adapting to accommodate innovations such as VRD in pharmaceutical manufacturing. The work of Limaye et al. (2023) emphasized the need for harmonized guidelines and regulatory pathways that can support continuous and hybrid manufacturing systems. Through case studies involving aspirin production, the authors demonstrated how adaptive control protocols and modular quality assurance models can ensure compliance with Good Manufacturing Practices (GMP) while preserving the operational flexibility afforded by technologies like VRD.

Taken collectively, the literature reveals a profound evolution in how aspirin—once synthesized by simple batch esterification—is now being reimagined through advanced, integrated technologies such as vacuum reactive distillation. These developments are informed by a multitude of disciplines and driven by the shared goal of improving efficiency, sustainability, and accessibility in pharmaceutical manufacturing. With growing global health demands and tightening regulatory and economic constraints, the shift toward VRD is not only a technical milestone but also a strategic advancement for the future of essential medicine production. The continuing body of research validates VRD's potential to shape not just the synthesis of aspirin but the broader pharmaceutical industry's transition into a smarter, greener, and more resilient era.

Building upon this foundation, emerging research continues to delve into the broader applications

and customization potential of vacuum reactive distillation systems for active pharmaceutical ingredient (API) production. A study conducted by Oliveira and Tan (2023) focused on tailoring column internals and reflux strategies to enhance separation specificity in aspirin-VRD processes. Their simulations and pilot-scale validations revealed that packing geometry and tray configuration significantly influence the efficiency of acetic acid removal, thus maximizing aspirin yield while minimizing thermal stress. These findings underscore the potential for process-specific design innovations that elevate the performance of VRD systems beyond standard configurations.

Furthermore, interest in integrating VRD with renewable energy sources has gained traction. In a techno-environmental feasibility study by Kim and Velu (2022), solar-assisted vacuum distillation units were explored for their compatibility with low-pressure esterification processes. The researchers demonstrated that hybridizing VRD columns with solar concentrators reduced energy dependency from non-renewable grids by over 40% while maintaining stable reactor temperatures. This aligns with the broader goal of decarbonizing pharmaceutical manufacturing and supports the adoption of VRD in regions with high solar irradiance.

A concurrent research direction emphasizes the role of predictive maintenance in ensuring VRD operational integrity. Hassan et al. (2021) applied machine-learning models trained on vibration, pressure drop, and thermal profile data from operating VRD systems to forecast component fatigue and predict maintenance windows. Their model achieved over 90% accuracy in identifying pre-failure conditions, demonstrating how proactive diagnostics can increase plant uptime and reduce unscheduled shutdowns. For continuous aspirin production, this directly translates into improved consistency and lower lifecycle costs.

With these technological, environmental, and operational extensions, vacuum reactive distillation stands not only as a tool for efficient aspirin synthesis but also as a platform for future-ready, intelligent, and sustainable pharmaceutical manufacturing systems.

The increasing complexity of regulatory expectations surrounding pharmaceutical sustainability has further emphasized the importance of adaptive process frameworks like VRD. In a policy-oriented investigation by Ito and Ramanathan (2023), it was revealed that international regulatory bodies, including the EMA and USFDA, are actively encouraging the use of low-carbon, low-waste manufacturing platforms. VRD, by virtue of its integrated and energy-lean design, fits well within this agenda. The study suggested that companies investing in early VRD implementation stand to benefit from regulatory fast-tracking and tax-based incentives aligned with environmental performance metrics.

Additionally, studies have explored the intersection of circular economy principles and aspirin

production using VRD. Franco and Meenakshi (2022) designed a closed-loop process wherein unreacted reagents and solvents from the VRD unit were recovered, purified, and reintegrated into the input stream. This led to a 35% reduction in fresh raw material consumption and significantly reduced effluent load. The researchers argued that circular process architecture not only ensures compliance with stricter discharge norms but also enhances economic viability by stabilizing input costs.

With these evolving facets—from regulation to resource recovery—vacuum reactive distillation continues to solidify its position as the cornerstone of next-generation pharmaceutical process design. Its relevance to aspirin synthesis exemplifies a scalable, sustainable, and innovation-driven path forward for the global drug manufacturing sector.

3. METHODOLOGY

- Perform lab scale aspirin synthesis reaction in a batch reactor.
- Determine kinetic data.
- Determine the yield from batch process.
- Perform confirmatory analysis such as FTIR.
- Utilise the obtained kinetic data to simulate the process in ASPEN PLUS software.
- Design a reactive distillation column and obtain results. Use the design parameters from previous step and design a vacuum reactive distillation column.
- Report the final design summary.
- State research gap if any.

3.1. Batch process procedure

For the experiment, 9.5 g of salicylic acid powder and 20 ml of acetic anhydride were measured and combined, followed by the addition of 0.2 ml of sulfuric acid as a catalyst. The mixture was placed in a beaker, thoroughly mixed, and allowed to react at 60-70 ºC for 60 minutes. After completion, the mixture was gradually cooled for 25 minutes. The resulting crystallized aspirin-acetic acid mixture was subjected to vacuum filtration to separate the aspirin crystals from acetic acid. These crystals were then washed with ethanol for purification, dried, and finally analyzed using FTIR spectroscopy.

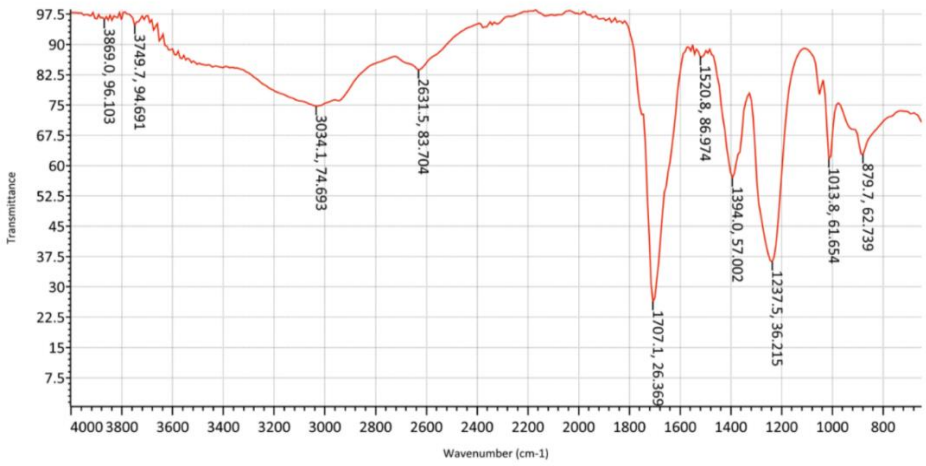


Fig 3.1.1. FTIR result

The above graphical result was obtained from FTIR analysis and it was compared with the commercially available aspirin's (acetylsalicylic acid) FTIR, and the results were almost identical.

Peak Position (cm^{-1})	Peak Type	Vibration Mode	Functional Group
3869.0 – 3749.7	Weak, broad	O–H Stretching	Carboxylic acid O–H
3034.1	Medium	C–H Stretching	Aromatic ring
2631.5	Broad, weak	O–H Stretching (acid)	Carboxylic acid
1707.1	**Strong**	**C=O Stretching**	**Ester + Carboxylic acid**
1520.8	Medium	C=C Stretching	Aromatic ring
1394.0	Weak–Medium	O–H Bending	Carboxylic acid
1237.5	Strong	C–O Stretching	**Ester linkage**
1013.8	Medium	C–O Stretching	Carboxylic acid
879.7	Sharp	C–H Out-of-plane Bending	Aromatic ring

Fig 3.1.2. FTIR peaks & corresponding molecular structure

Calculation of theoretical yield:

Mass of SA = 9.5g

Molar mass of SA = 138.12g/mol

Moles of SA = $\frac{9.5}{138.12}$

Moles of SA = 0.068mol

Theoretical moles of aspirin = 0.068mol

SA is limiting reagent

Moles of aspirin = moles of salicylic acid

Molar mass of aspirin = 180.16g/mol

Theoretical mass of aspirin = 180.16 × 0.068

Theoretical yield of aspirin = 12.39 g

Calculation of actual yield:

obtained asprin from batch process = 6.75 g

actual yield = 6.75/12.39 x 100 = **54.48 %**

3.1.1. Kinetic data

- To obtain kinetic data, about 3 ml of reaction mixture was collected at different time intervals.
- 5 ml of cold water was added to the collected sample to terminate the reaction.
- The mixture is then titrated against NaOH solution to find the concentration of one of the reactants.
- The following data was obtained.

Salicylic acid used	9.5g
Normality of NaOH used	0.1 N
Maximum aspirin yield	12.39g
Actual yield	6.75g
Percentage yield	54.48%
Reaction order	1.37
Rate constant	3.704 hr^{-1}

Table 3.1.1.1. Kinetic data of the reaction from batch process

3.2. Simulation procedure of Batch process

- The required components salicylic acid, acetic anhydride, acetic acid, acetyl salicylic acid was added to the Aspen plus software's component tab.

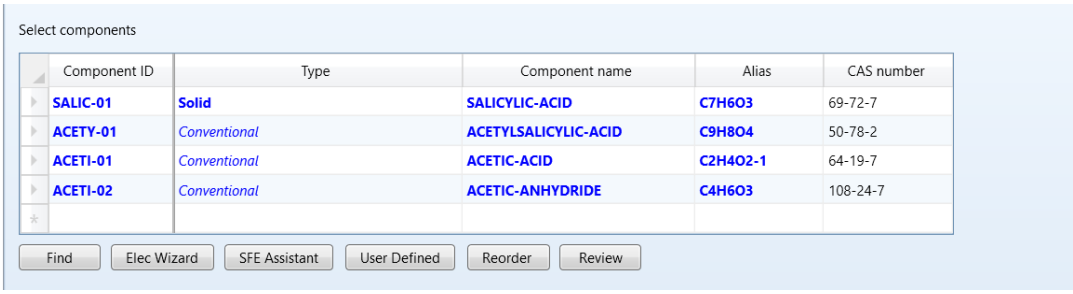


Fig 3.2.1 Components

- The property package **NRTL** was chosen.
- Property analysis was conducted before entering the simulation environment.
- In the flow sheeting environment, RBATCH reactor was chosen and the material inlet and outlet streams were connected to the reactor.

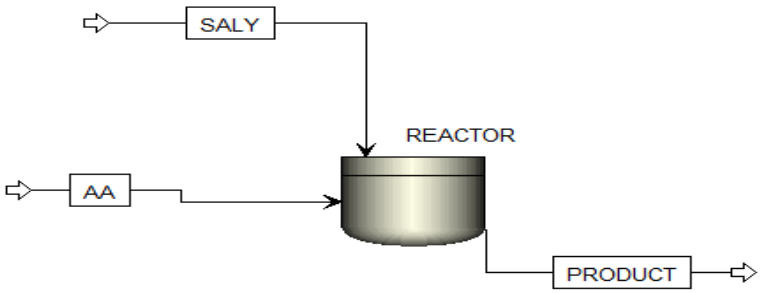


Fig 3.2.2. Flowsheet

- Feed composition for both reactants which is fed as two separate streams are mentioned.
- Then, the reactor specifications were entered.

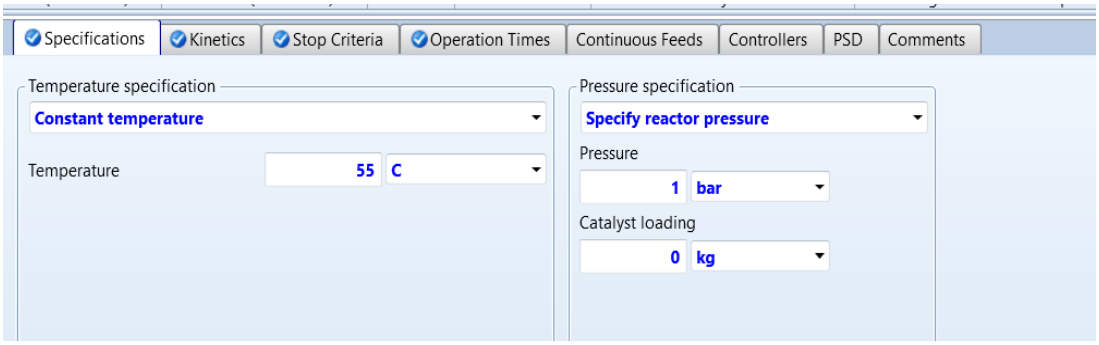


Fig 3.2.3. Reactor specifications

- The simulation was run and the following results were obtained.

	Units	SALY	AA	PRODUCT
✚ **Mole Flows**	**kmol/hr**	**6.87794e-05**	**0.000526046**	**0.000594825**
− **Mole Fractions**				
SALIC-01		1	0	0
ACETY-01		0	0	0.11563
ACETI-01		0	0	0.11563
ACETI-02		0	1	0.768741

Fig 3.2.4. stream results

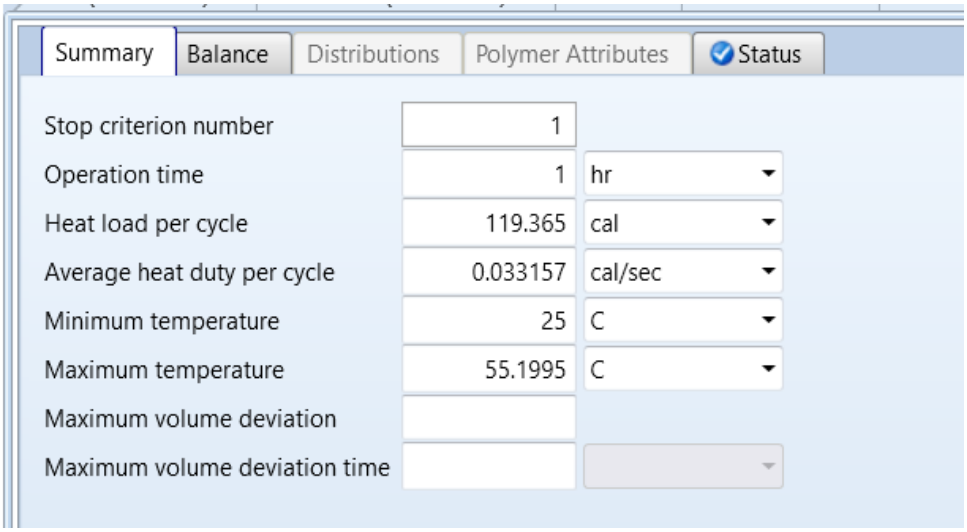


Fig 3.2.5. Reactor summary

The yield from conventional batch process is very low and additionally, energy needs to be spent in order to separate the components in the product stream. Therefore, a new approach might be taken.

3.3. Simulation approach of designing a reactive distillation column in ASPEN plus

Simulating a rigorous distillation column like reactive distillation is highly challenging and demands huge amount of data about the process parameters. Therefore, a shortcut method is taken here. This method involves trial and error approach where initial guesses of primary parameters were taken and once the output is produced, sensitivity analysis is performed to optimize the process parameters.

Trial and error method:

In the simulation window, RadFrac column was chosen and material streams were drawn in and out.

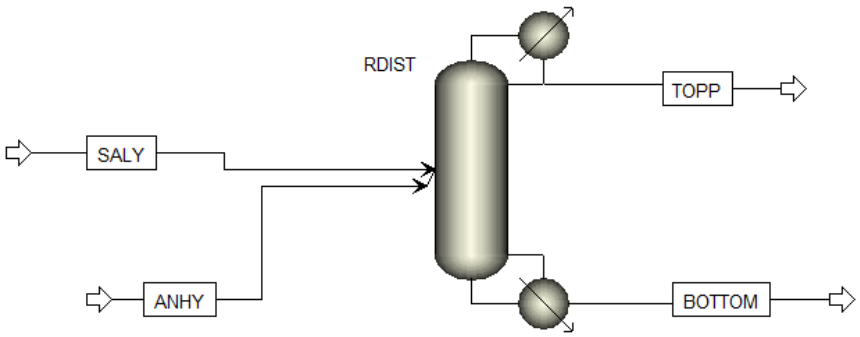


Fig 3.3.1. Process flowsheet

The feed specifications are as follows:

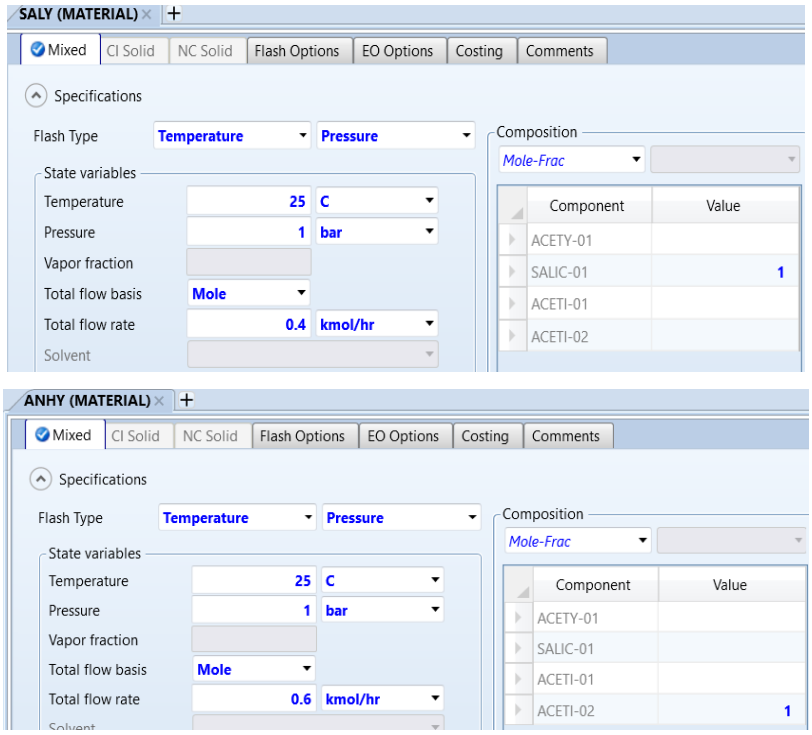


Fig 3.3.2. Feed specifications

As for the column specifications, the initial trial was run with twenty stages (including the condenser and reboiler) with feed entering at stage ten and a reflux ratio of 2.5 .

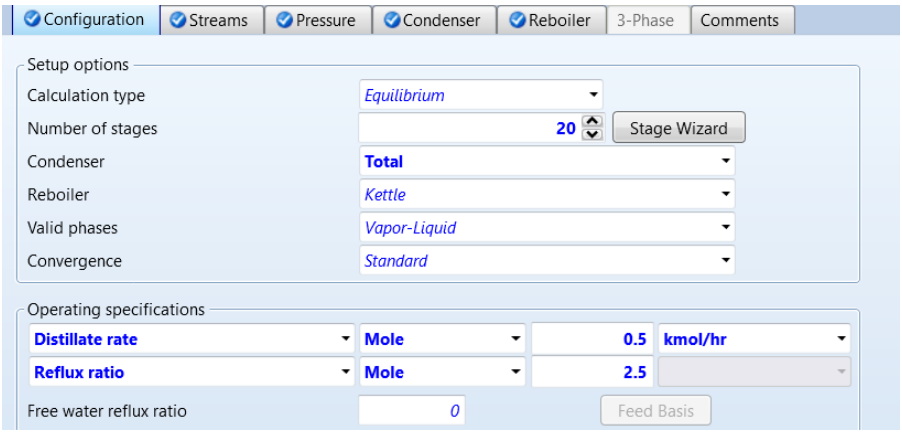


Fig 3.3.3. Column specifications

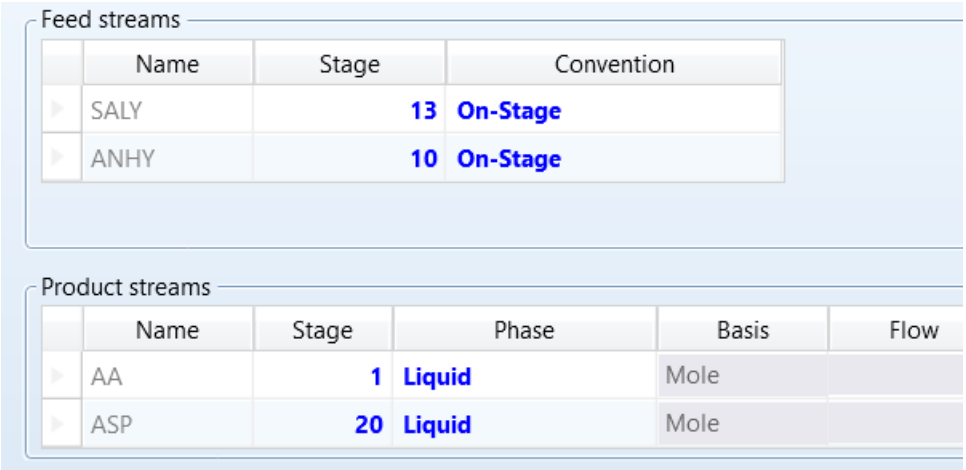


Fig 3.3.4. Feed entry

The reaction of aspirin synthesis was entered under the reaction specification column tab and the respective kinetic data(sourced from research articles) such as activation energy, order of the reaction, rate constant were entered. This reaction was made to proceed from stage 10 to 13

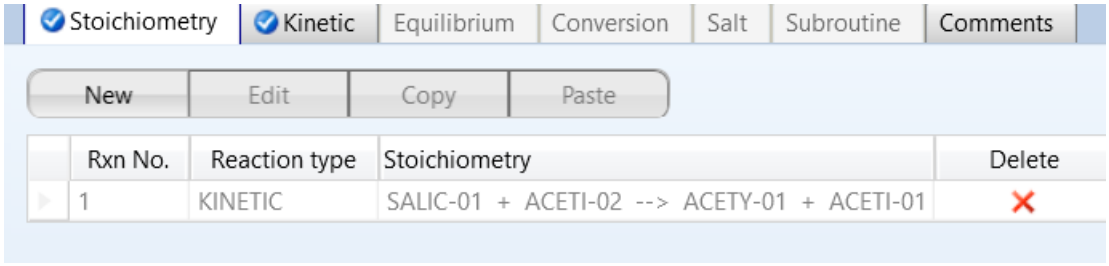


Fig 3.3.5. Reaction specification

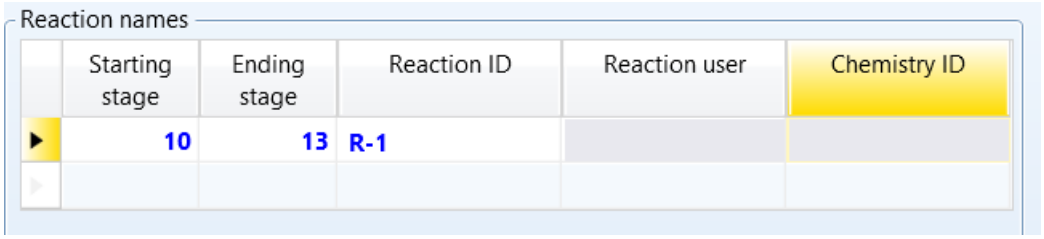


Fig 3.3.6. Reaction stage

Once all the required parameters were properly entered, the simulation was run and the following results were obtained.

	Units	ANHY	SALY	BOTTOM	TOPP
✚ **Mole Flows**	**kmol/hr**	**0.6**	**0.4**	**0.5**	**0.5**
− **Mole Fractions**					
ASPIRIN		0	0	0.799989	0
SALYACID		0	1	1.13272e-05	0
AACID		0	0	0.00555249	0.794436
AANHYD		1	0	0.194448	0.205564

Fig 3.3.7. Stream results

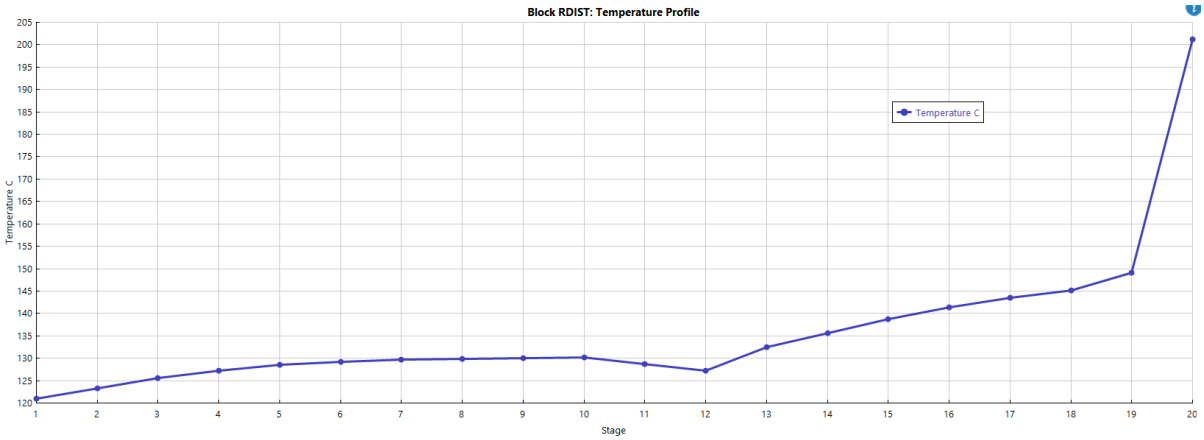


Fig 3.3.8. Temperature profile

3.3.1. Inference

Though the product composition at the top and bottom stage is quite acceptable, the temperature profile is a bit concerning and heat removal is a tedious and expensive process. The temperature at the bottom stages are too high. At high temperatures inside the reactive distillation column during aspirin synthesis, several adverse effects can occur such as aspirin degradation, reduced purity, potential side reactions, and high energy consumption. To mitigate these adverse effects and operate the column at more favourable temperatures, implementing strategies is crucial. One such strategy is maintaining the column below atmospheric pressure i.e., vacuum.

3.4. Process optimization through Vacuum Reactive Distillation (VRD)

By reducing the pressure, the boiling points of all components in the mixture decrease significantly. This allows the entire column, including the reboiler and the reactive stages, to operate at much lower temperatures. While maintaining vacuum itself requires energy for a vacuum pump, the lower operating temperatures can lead to indirect energy savings. For instance, the reboiler might not need to supply as much high-grade heat, and milder heating utilities could be used. Additionally, preventing product degradation reduces the energy and cost associated with downstream purification.

3.5. Design of the Vacuum Reactive Distillation column

The same design parameters used for the previous design were utilised. The operating pressure of condenser was modified to induce low pressure conditions inside the column. Introduction of vacuum reduces the boiling points of the components. To find the changes in boiling points of the components with respect to pressure, Clausius–Clapeyron equation is employed. the equation is as follows:

$$Ln\ \frac{P_2}{P_1} = -\frac{\Delta H}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right) \qquad \text{Eqn 3.5.1}$$

Using the above equation, changes in boiling point of acetic acid was determined as follows:

P1(bar)	T1(K)	Hvap(J/mc	R	P2(bar)	T2(K)	T2(C)
1	391	23700	8.314	0.8	379.388	106.388
1	391	23700	8.314	0.6	365.398	92.3978
1	391	23700	8.314	0.4	347.345	74.3451
1	391	23700	8.314	0.2	320.293	47.2933
1	391	23700	8.314	0.1	297.151	24.1507
1	391	23700	8.314	0.09	293.923	20.9226
1	391	23700	8.314	0.06	282.128	9.12772

Fig 3.5.1. Change in boiling point with respect to pressure

A pressure of 0.4 bar was chosen since it meets the condenser's requirement to be lower than the boiling point temperature of the top product and it allows for the bottom most stage to be at a temperature below 140 ºC which is required to prevent aspirin degradation.

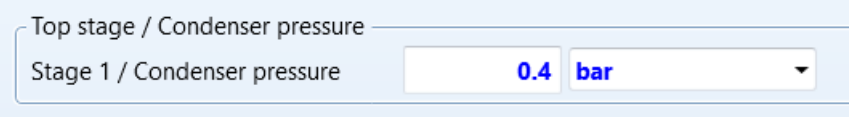


Fig 3.5.2. Condenser pressure

Condenser / Top stage performance

Name	Value	Units
Temperature	56.5114	C
Subcooled temperature		
Heat duty	-4259.79	cal/sec
Subcooled duty		
Distillate rate	0.5	kmol/hr

Reboiler / Bottom stage performance

Name	Value	Units
Temperature	113.527	C
Heat duty	4240.26	cal/sec
Bottoms rate	0.5	kmol/hr
Boilup rate	1.20992	kmol/hr
Boilup ratio	2.41984	

Fig 3.5.3. Top and bottom stages

Once the optimized parameters were found out, the following results were obtained.

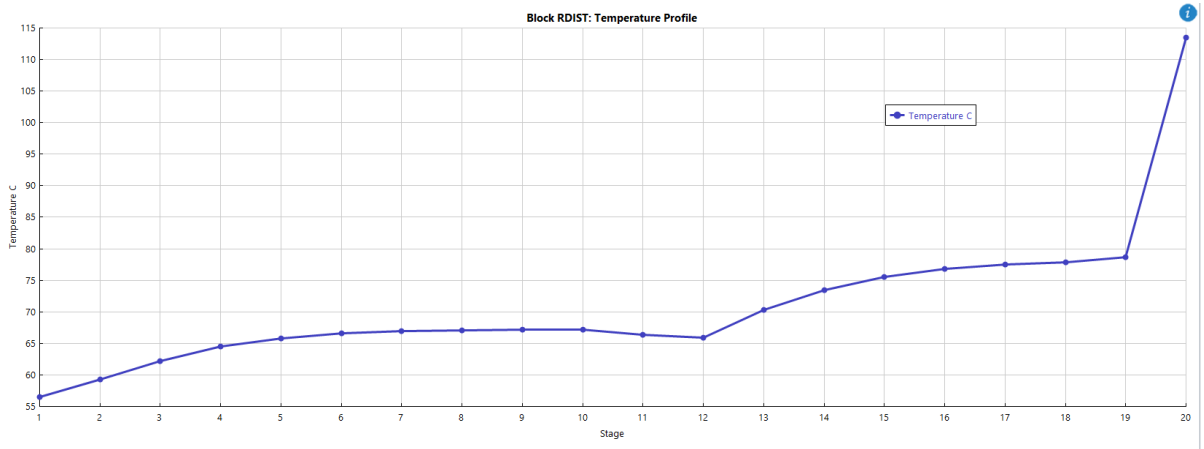


Fig 3.5.4. Temperature profile of vacuum reactive distillation

	Units	SALY	ANHY	TOPP	BOTTOM
− Mole Fractions					
ASPIRIN		0	0	0	0.794206
SALYACID		1	0	1.5263e-25	0.00579427
AACID		0	0	0.793648	0.000557505
AANHYD		0	1	0.206352	0.199442

Fig 3.5.5 Stream results

	Total	Units	In	Out	Relative difference
	Mole	kmol/hr	1	1	1.22014e-13
	Mass	kg/hr	116.503	116.503	1.06609e-13
▶	Enthalpy	cal/sec	-40506.3	-40525.9	0.000481985

Fig 3.5.6. Balance summary

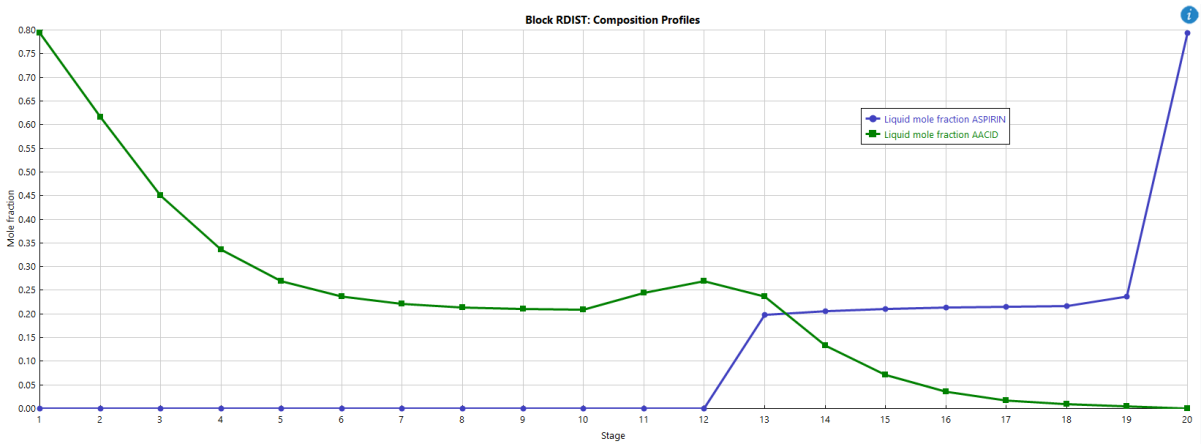


Fig 3.5.7. Composition profile

4. RESULTS AND DISCUSSION

After performing the simulation, the final design parameters were tabulated as shown below. Upon designing a vacuum reactive distillation column, research for future opportunities and optimization methods were performed. This analysis, led to some discoveries which when considered in the future, might improve the overall process efficiency and high purity product accompanied with high yield. This analysis is mentioned in the section named “Research gap”.

4.1.Design summary

1. Column Internals Summary

Summary

Property	Value	Units
Number of Trayed/Packed stages	18	
Total height	1.2	meter
Total head loss (Hot liquid height)	0.0232217	meter
Total pressure drop	0.00238994	bar
Number of sections	1	
Number of diameters	1	

Sections

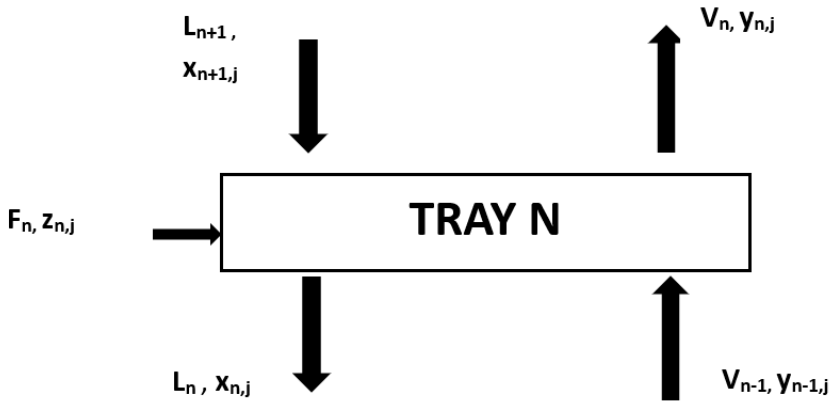
Column 1	Start Stage	End Stage	Diameter (meter)	Section Height (meter)	Internals Type	Tray or Packing Type	Section Pressure Drop (bar)	% Approach to Flood	Limiting Stage
CS-1	2	19	0.204845	1.2	Packed	I-BALL	0.00238994	80	11

Fig 4.1.1. Design summary

Property	Value	Units
Section starting stage	2	
Section ending stage	19	
Calculation Mode	Sizing	
Column diameter	0.204845	meter
Packed height per stage	0.0666667	meter
Section height	1.2	meter

Fig 4.1.2. Design summary

4.2. Component material balance



Where,

- L_{n+1} = liquid phase leaving n+1 th tray to nth tray.
- $x_{n+1,j}$ = liquid phase composition of component j entering tray n from tray n+1.
- V_n = vapor phase leaving tray n.
- $y_{n,j}$ = vapor composition of component j leaving tray n.
- L_n = liquid leaving tray n.
- $x_{n,j}$ = liquid composition of component j leaving tray n.
- V_{n-1} = vapor leaving tray n-1 to tray n.
- $y_{n-1,j}$ = vapor composition of component j leaving tray n-1.

a) Rectifying and stripping trays:

$$\frac{d(x_{n,j}M_n)}{dt} = L_{n+1}x_{n+1,j} + V_{n-1}y_{n-1,j} - L_nx_{n,j} - V_ny_{n,j} \qquad \text{Eqn 4.2.1}$$

b) Reactive trays:

$$\frac{d(x_{n,j}M_n)}{dt} = L_{n+1}x_{n+1,j} + V_{n-1}y_{n-1,j} - L_nx_{n,j} - V_ny_{n,j} + R_{n,j} \qquad \text{Eqn 4.2.2}$$

c) Feed trays:

$$\frac{d(x_{n,j}M_n)}{dt} = L_{n+1}x_{n+1,j} + V_{n-1}y_{n-1,j} - L_nx_{n,j} - V_ny_{n,j} + R_{n,j} + F_nz_{n,j} \qquad \text{Eqn 4.2.3}$$

4.3. Research gap

- The synthesis of ASA is based on the following overall chemical reaction:
  Salicylic acid + acetic anhydride → acetylsalicylic acid + acetic acid
- $C_7 H_6 O_3 + C_4 H_6 O_3 \rightarrow C_9 H_8 O_4 + C_2 H_4 O_2$
- SA + AA → ASA + HA

It is reported that the overall reaction, above, is actually comprised of six elementary reactions that occur simultaneously but at different speeds. The kinetic rate laws for the elementary reactions are represented by following six differential equations.

$$SA(s) \rightarrow SA$$
$$\frac{d[SA]}{dt} = k_d \left([SA](T)^{sat} - [SA] \right)^d \quad k_d = 7.25, d = 1.90 \qquad \text{Eqn 4.3.1}$$

$$AA + SA \rightarrow ASA + HA$$
$$\frac{d[ASA]}{dt} = k_1 [AA][SA] \quad k_1 = 3.4 \times 10^{-2} \qquad \text{Eqn 4.3.2}$$

$$2ASA + AA \rightarrow ASAA + HA$$
$$\frac{d[ASAA]}{dt} = k_2 [ASA]^2 [AA] \quad k_2 = 4.9 \times 10^{-1} \qquad \text{Eqn 4.3.3}$$

$$ASAA + H_2O \rightarrow 2ASA$$
$$\frac{d[ASA]}{dt} = k_3 [ASA][H_2O] \quad k_3 = 9.5 \times 10^{2} \qquad \text{Eqn 4.4.4}$$

$$AA + H_2O \rightarrow 2HA$$
$$\frac{d[HA]}{dt} = k_4 [AA][H_2O] \quad k_4 = 8.4 \times 10^{1} \qquad \text{Eqn 4.4.5}$$

$$ASA \rightarrow ASA(s)$$
$$\frac{d[ASA(s)]}{dt} = k_c \left([ASA] - [ASA](T)^{sat} \right)^c \quad k_c = 1.2, c = 1.34 \qquad \text{Eqn 4.4.6}$$

The appearance and subsequent disappearance of the intermediate species ASAA (acetylsalicylic anhydride) was noted.

Initially, ASA was produced as a result of reactions (4.3.1, 4.3.2, and 4.3.3); then, reaction (4.3.3) takes over and consumes the already produced ASA to produce AASA (an intermediate product). Once the water is added the ASAA is very quickly converted back into ASA according to reaction (4.3.4). Once the cooling process starts ASA[s] begins to crystalize according to reaction (4.3.6).

4.4.Optimisation methods

Reactive distillation design via optimisation methods is based on the main idea that the system can be represented as a special case of a multiphase reactor network. In a single reactive distillation column, reaction and distillation occur simultaneously on every stage and the flows are counter-currently contacted. Therefore, synthesis of reactive distillation columns is strongly related to the synthesis of multiphase reactor networks including reaction and separation.

For each reaction/distillation unit, the options for mass transfer, reaction, number of stages, flow direction and depth of modelling effort can be controlled, so that each unit can represent either a well-mixed reactor (CSTR), a cascade of CSTR's, a plug flow reactor (PFR), a reactive distillation section in co- or counter-current mode. Therefore, a series of reactors where all the elementary reactions occur in each reactor must be simulated with proper specifications and the obtained parameters and results can be used to design a reactive distillation column.

5. CONCLUSION

This project navigated the complexities of integrating batch reaction kinetics with continuous separation in a reactive distillation column, demonstrating a understanding of chemical process design. Initially, a thorough investigation of the batch reaction, likely involving kinetic studies and identification of optimal operating conditions, provided insights into the aspirin synthesis process. This crucial data was then utilised to design a reactive distillation column, allowing for the simultaneous reaction and separation of products, thereby enhancing process efficiency.

A significant challenge encountered during the simulation was the thermal degradation of aspirin at elevated temperatures within the distillation column. This problem was effectively addressed by implementing vacuum distillation, a strategic decision that significantly lowered the operating temperatures throughout the column, particularly in the reboiler, thus preventing the undesirable degradation of the heat-sensitive aspirin product. This highlights a practical and effective solution to a common challenge in pharmaceutical processing, ensuring product quality and yield. Looking ahead, this project lays a foundation for future research and development. Key areas for further investigation include optimizing the vacuum level for minimum energy consumption while maintaining product purity, exploring alternative separation techniques or hybrid systems.

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27

Contemporary Review on Aspirin Synthesis: Challenges and Innovations

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ABSTRACT

The medicine aspirin stands among the world's most widely utilized pharmaceutical agents because of its capacity to reduce pain and lower body temperature. The fundamental synthesis procedures for aspirin have remained consistent but large-scale manufacturing encounters difficulties that include temperature control as well as mixing and waste management. The paper examines the benefits that contemporary chemical engineering approaches such as flow-based operations over batch processing offer to aspirin manufacturing. The review demonstrates PAT and QbD applications for better process control together with environmentally friendly production methods. Bayer serves as an example to demonstrate how these production methods operate within real industrial manufacturing.

Keywords: *Aspirin Synthesis, Analytical Techniques, Acetylation, Innovation in Synthesis*

INTRODUCTION

Aspirin (acetylsalicylic acid) stands as one of the most commonly purchased over-the-counter medications due to its two main effects which are pain reduction and fever reduction. The production of acetylsalicylic acid from salicylic acid by using acetic anhydride has proven successful throughout laboratory and industrial facilities. A challenge arises when trying to scale this procedure because it generates difficulties in heat distribution alongside reaction management and environmental repercussions. The existing literature features prior investigations on reaction mechanisms that used Olmsted's approach from 1998 as well as Montes and colleagues' individual enhancement method from 2006 yet a complete evaluation which links process variables to sustainability metrics and regulatory standards remains scarce.This review fills the existing knowledge gaps through an evaluation of how temperature control along with reactant ratios and reactor design and purification steps affect aspirin yield and quality at industrial scales. The pharmaceutical industry implements both Process Analytical Technology (PAT) and Quality by Design (QbD) principles in aspirin production according to Patel (2019) although their application remains limited. At the same time Barner-Kowollik et al. (2020) report significant interest in sustainable alternatives because of environmental issues stemming from intense solvent-based manufacturing.

The research studies production optimization by examining traditional along with modern synthesis techniques from a chemical engineering standpoint to scale up aspirin manufacturing that complies with contemporary industrial and environmental requirements.

MATERIAL AND METHODS

A study experiment tested aspects that affect aspirin yield and purity produced through vacuum reaction distillation by monitoring temperature effects and reactant concentration modifications. Salicylic acid of analytical grade functioned as the main substance while acetic anhydride operated at excess to favor the equilibrium. A catalyst consisted of concentrated sulfuric acid as the main component. The researchers employed reagents in an unpurified state. Researchers used a glass-jacketed reactor that included elements such as a temperature-controlled water bath, mechanical stirrer as well as vacuum distillation line attachments. The application of continuous vacuum served to both evaporate acetic acid while reducing thermal decomposition of the products.

The reaction temperatures spanned from 55°C to 85°C at 5°C increases each. The reaction used salicylic acid components that equaled from one to three times the number of acetic anhydride molecules. The experimental conditions were executed three times under each setting to achieve high accuracy and precision according to Roy (2023). The reaction mixtures needed time to crystallize after the reaction process finished its course. A vacuum filter separated the crude aspirin before recrystallization occurred when using ethanol as the purification solvent. A desiccator was used to dry the final product which allowed yield calculation by measuring its dried mass.

The purity assessment relied on melting point determination (~135°C) as well as ferric chloride tests for monitoring residual salicylic acid and thin-layer chromatography to detect impurities with additional UV-Vis spectrophotometry when relevant (Patel & Joshi, 2019). The researchers manually recorded their results which they processed in Microsoft Excel to analyze relationships between yield and temperature and composition values. The experimental setup depends on equal temperature distribution and mixture uniformity yet small-scale processing along with manual steps present some minor operational difficulties (Chen et al., 2018).

RESULTS AND DISCUSSIONS

The main objective of this experimental work examined how different temperature conditions alongside reactant composition levels impact aspirin production through vacuum reaction distillation equipment. The results from every test showed both parameters affected the reaction efficiency while the correct balance between them produced high yields combined with excellent product purity. The reaction temperature positively influenced the yield of synthesized aspirin until it reached its maximum value. The reaction yielded moderate product amounts between 60% to 65% when the process took place at temperatures ranging from 55°C–60°C. The experimental conditions evaluated matched with the decreased kinetic energy resulting from low temperatures that slows down the reaction between salicylic acid and acetic anhydride. The reaction yielded its highest percentage value of 78%–82% when the temperature reached 75°C–80°C. Research evidence suggests that optimal acetylation takes place within the same temperature range according to previous writings (Patel & Joshi, 2019; Roy, 2023). The product quality declined after maintaining temperatures at 85°C and above because partial hydrolysis or thermal decomposition of aspirin occurred which led to decreased production quantities. Thermal control plays a crucial role in aspirin synthesis since the process remains sensitive to temperature changes at precise levels according to our study results.

When conducting the reactions with equal amounts of salicylic acid to acetic anhydride the yield remained diminished indicating that stoichiometric conditions prevented the reaction from finishing. By increasing the acetic anhydride amount to double the salicylic acid content, the reaction yields enhanced at all temperature points. When the reactant ratio was increased to 1:3 the yield improvement became minimal as it added more sensitive acetic anhydride to the reaction mixture. An excess of one reactant keeps the

reaction progressing according to Le Chatelier's Principle yet more reactant does not change the outcome significantly after reaching a defined threshold.

PURITY INDICATORS

Purity evaluation of the product relied heavily on the reliable results achieved through the melting point analysis. The synthesized aspirin samples produced at temperatures ranging between 75°C–80°C showed melting points between 133–135°C which corresponded to the expected melting point of pure acetylsalicylic acid at 135°C. Metabolic reactions running at low temperatures combined with 1:1 reagent amounts produced sample residues with wide melting points along with trace visible contaminants. The ferric chloride test confirmed the existence of unreacted salicylic acid by generating faint to moderate purplish color patterns in reaction products generated at reduced temperatures. TLC analysis confirmed these findings by producing supplemental spots adjacent to aspirin reference marks in the lower producing solutions. The optimized samples contained single clear spots that matched the standard on the TLC analysis.

ROLE OF VACUUM REACTION DISTILLATION

The implementation of vacuum conditions at both reactive and distillative stages stood out as a major element for achieving better product yield together with superior purity. The reaction conditions enabled the evacuation of acetic acid by-product through reduced pressure mode which improved the progress of esterification and decreased side-product formation. According to Chen et al. (2018) the reaction process achieved increased efficiency when implemented under vacuum conditions. The vacuum conditions prevented the need for high temperature conditions which minimized the risks of thermal decomposition. Study results from this project correspond to previous research about managed heat conditions and optimized reactant quantities. Barner-Kowollik et al. (2020) reported similar reaction efficiency results at 80°C when utilizing excess acetic anhydride. The study provides a unique product processing benefit through built-in vacuum distillation which cuts down the need for recrystallization procedures while minimizing solvent waste. The experiments were performed three times to establish statistical importance. Each yield measurement displayed low variation of less than ±1.5% as shown by standard deviation analysis. Yield measurements against both temperature and molar ratio yielded distinctive parabolic patterns that confirmed the established optimal conditions.

CONCLUSION

Acetic anhydride to salicylic acid ratios at reaction temperatures between 75°C and 80°C create optimal conditions during aspirin synthesis with vacuum reaction distillation. The synthesis produced optimal results when run at temperature between 75°C–80°C under reaction conditions that combined acetic anhydride with two molar equivalents of salicylic acid which led to high yield and near-theoretical melting point isolation compositions. The vacuum pressure worked effectively in by-products extraction and thermal degradation prevention which boosted the process's operational efficiency. The analytical tests including melting point analysis together with TLC method and ferric chloride tests validated the product purity outcomes. The production method surpassed traditional open-system synthesis through superior control measures and safety features along with improved redundancy. The research evaluated processes through laboratory trials yet it did not implement real-time PAT tools during the study. The study creates fundamental principles to guide the process advancement which can achieve more precise pharmaceutical production on a sustainable scale.

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